# Linear Flow: Diffusivity Equation Derivation

## Notation Guide

Every derivative in this PDF is written out explicitly. The compact symbol ∇ (“del”) is not used.

| Symbol | Meaning |
|---|---|
| ∂p/∂x | how fast pressure changes with x, with the other variables held fixed (the slope of p in the x direction) |
| $\partial^2 p/\partial x^2$ | the slope of the slope (curvature of p in x) |
| ∂p/∂t | how fast pressure changes with time |
| [ρv] at x | the value of the product ρv evaluated at position x |
| Δx, Δt | small increments of position and time |
| η = k/(φμc_t) | hydraulic diffusivity |
| c, c_f, c_t | fluid, rock (formation) and total compressibility |
| p′ | pressure drawdown (used in the gravity section) |

Convention: flow in the **−x direction**, mass balanced over a time interval **Δt**. Every step is written out.

## Part 0: Setup and assumptions

**Geometry:** a rectangular block of porous rock with width w, height h, and flow only along x. The cross-sectional area is

$$A = w\,h$$

We study a thin slice between x and x + Δx. Its bulk volume is

$$\Delta V = w\,h\,\Delta x$$

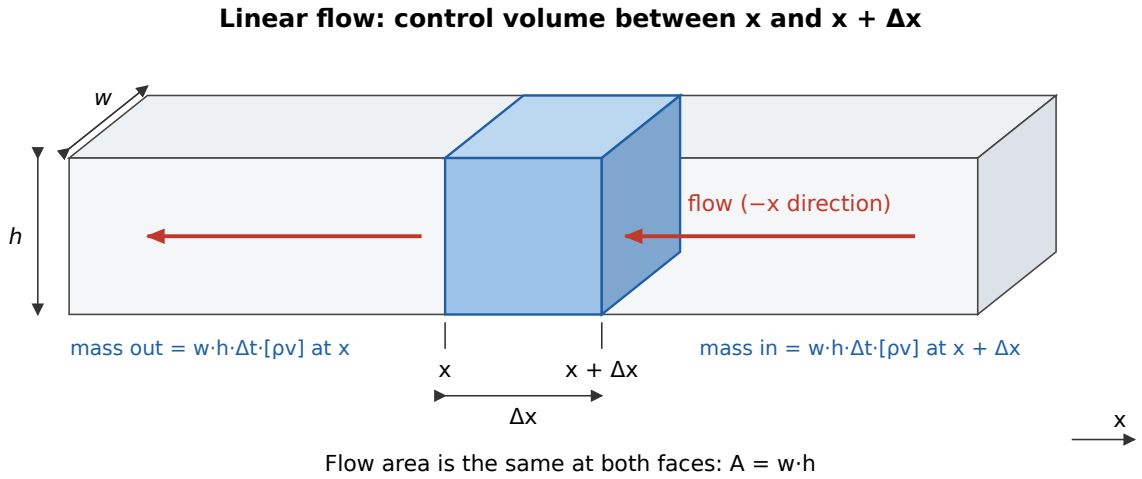



Figure 1. Linear flow. Fluid enters the slice at x + Δx and leaves at x. The flow area w·h is the same on both faces.

**Assumptions:**

1. Flow in one direction only (x)
2. Single-phase, slightly compressible fluid
3. Homogeneous rock: k constant. Fluid viscosity μ constant
4. Darcy's law applies
5. Horizontal flow (no gravity)
6. Isothermal
7. Small pressure gradients

**Variables:**

- ρ = fluid density
- v = Darcy velocity (volume flow rate per unit area)
- φ = porosity
- p = pressure
- c = fluid compressibility
- c_f = rock (pore) compressibility

# Part 1: Mass of fluid crossing a face

Mass = density × volume. The volume of fluid crossing area A at velocity v during time Δt is A·v·Δt. Multiply by density:

$$\text{mass} = \rho\, A\, v\, \Delta t = \rho\, v\, w\, h\, \Delta t$$

**Units check:** (kg/m³)·(m/s)·m²·s = kg ✓

# Part 2: Mass in, mass out, mass accumulated

Fluid moves in the −x direction. It **enters** the slice through the right face at x + Δx and **leaves** through the left face at x.

**Mass in** (through the face at x + Δx):

$$(\text{ mass })_{in} = w\,h\,\Delta t\ [\,\rho v\,]_{x+\Delta x}$$

**Mass out** (through the face at x):

$$(\text{ mass })_{out} = w\,h\,\Delta t\ [\,\rho v\,]_{x}$$

**Mass accumulated** inside the slice. The mass stored in the slice at any time is:

$$\text{mass stored} = (\text{ pore volume }) \times \rho = (\,\phi\,\Delta V)\ \rho = \Delta V\,(\,\phi\rho\,)$$

The accumulation is the change in stored mass between time t and t + Δt:

$$(\text{ mass })_{acc} = \Delta V[\,(\,\phi\rho\,)_{t+\Delta t} - (\,\phi\rho\,)_{t}]$$

# Part 3: Mass balance

Conservation of mass says matter is neither created nor destroyed:

$$(\text{ mass })_{in} - (\text{ mass })_{out} = (\text{ mass })_{acc}$$

Substitute the three expressions:

$$w\,h\,\Delta t\ [\,\rho v\,]_{x+\Delta x} - w\,h\,\Delta t\ [\,\rho v\,]_{x} = w\,h\,\Delta x\,[\,(\,\phi\rho\,)_{t+\Delta t} - (\,\phi\rho\,)_{t}]$$

Factor w·h·Δt on the left:

$$w\,h\,\Delta t\,(\,[\,\rho v\,]_{x+\Delta x} - [\,\rho v\,]_{x}) = w\,h\,\Delta x\,[\,(\,\phi\rho\,)_{t+\Delta t} - (\,\phi\rho\,)_{t}]$$

# Part 4: Divide and take limits

Divide both sides by w·h·Δx·Δt.

- Left side: w·h cancels, Δt cancels, leaving Δx in the denominator.
- Right side: w·h cancels, Δx cancels, leaving Δt in the denominator.

$$\frac{[\,\rho v\,]_{x+\Delta x} - [\,\rho v\,]_{x}}{\Delta x} = \frac{(\,\phi\rho\,)_{t+\Delta t} - (\,\phi\rho\,)_{t}}{\Delta t}$$

This is where Δt and w·h disappear from the equation.

Now let Δx → 0 and Δt → 0. By the definition of the derivative:

$$\lim_{\Delta x \to 0} \frac{f\left(\, x + \Delta x \,\right) - f\left(\, x \,\right)}{\Delta x} = \frac{\partial f}{\partial x}$$

So the left side becomes ∂(ρv)/∂x and the right side becomes ∂(φρ)/∂t. This is the continuity equation, eq. (1):

$$\frac{\partial\left(\, \rho v \,\right)}{\partial x} = \frac{\partial\left(\, \phi \rho \,\right)}{\partial t} \qquad (\, 1 \,)$$

# Part 5: Darcy's law

Darcy's law says flow goes from high pressure to low pressure at a rate proportional to the pressure gradient:

$$v = -\frac{k}{\mu}\frac{\partial p}{\partial s}$$

where s is the direction of flow. Here the flow is in the **−x direction**, so s = −x and ∂/∂s = −∂/∂x:

$$v = -\frac{k}{\mu}\left(-\frac{\partial p}{\partial x}\right) = +\frac{k}{\mu}\frac{\partial p}{\partial x} \qquad (\, 2 \,)$$

**Sanity check:** if pressure increases with x, then ∂p/∂x > 0 and v > 0. Fluid moves from high pressure (large x) toward low pressure (small x), which is the −x direction ✓

# Part 6: Combine continuity and Darcy

Substitute (2) into the left side of (1):

$$\rho v = \rho\frac{k}{\mu}\frac{\partial p}{\partial x}$$

$$\frac{\partial}{\partial x}\left(\rho\frac{k}{\mu}\frac{\partial p}{\partial x}\right) = \frac{\partial\left(\, \phi\rho \,\right)}{\partial t}$$

Since k and μ are constants, they can be pulled outside the derivative:

$$\frac{k}{\mu}\frac{\partial}{\partial x}\left(\rho\frac{\partial p}{\partial x}\right) = \frac{\partial\left(\, \phi\rho \,\right)}{\partial t} \qquad (\, 3 \,)$$

Now we must deal with ρ and φ, which both depend on pressure. For that we need compressibility.

# Part 7: Compressibility definitions

**Fluid compressibility:** the fractional change in density per unit pressure change:

$$c = \frac{1}{\rho}\frac{\partial \rho}{\partial p}$$

Multiply both sides by ρ:

$$\frac{\partial \rho}{\partial p} = c\,\rho \qquad (4)$$

**Rock compressibility:** the fractional change in porosity per unit pressure change:

$$c_f = \frac{1}{\phi}\frac{\partial \phi}{\partial p}$$

$$\frac{\partial \phi}{\partial p} = c_f\,\phi \qquad (5)$$

**Chain rule.** Since ρ and φ depend on x and t only through p(x,t):

$$\frac{\partial \rho}{\partial x} = \frac{\partial \rho}{\partial p}\frac{\partial p}{\partial x} = c\,\rho\,\frac{\partial p}{\partial x} \qquad (6)$$

$$\frac{\partial \rho}{\partial t} = \frac{\partial \rho}{\partial p}\frac{\partial p}{\partial t} = c\,\rho\,\frac{\partial p}{\partial t} \qquad (7)$$

$$\frac{\partial \phi}{\partial t} = \frac{\partial \phi}{\partial p}\frac{\partial p}{\partial t} = c_f\,\phi\,\frac{\partial p}{\partial t} \qquad (8)$$

# Part 8: Right-hand side of (3) (accumulation term)

Apply the product rule, ∂(ab)/∂t = a ∂b/∂t + b ∂a/∂t:

$$\frac{\partial\,(\,\phi\rho\,)}{\partial t} = \phi\,\frac{\partial \rho}{\partial t} + \rho\,\frac{\partial \phi}{\partial t}$$

Substitute (7) and (8):

$$= \phi\,(c\,\rho\,\frac{\partial p}{\partial t}) + \rho\,(c_f\,\phi\,\frac{\partial p}{\partial t})$$

Factor out φ ρ ∂p/∂t:

$$= \phi\,\rho\;(\,c + c_f\,)\;\frac{\partial p}{\partial t}$$

Define **total compressibility** c_t = c + c_f:

$$\frac{\partial\,(\,\phi\rho\,)}{\partial t} = \phi\,\rho\,c_t\,\frac{\partial p}{\partial t} \qquad (\,9\,)$$

# Part 9: Left-hand side of (3) (flow term)

Apply the product rule to ρ ∂p/∂x, treating ρ as a and ∂p/∂x as b:

$$\frac{\partial}{\partial x}\left(\rho\,\frac{\partial p}{\partial x}\right) = \rho\,\frac{\partial^2 p}{\partial x^2} + \frac{\partial p}{\partial x}\,\frac{\partial \rho}{\partial x}$$

Substitute (6), ∂ρ/∂x = c ρ ∂p/∂x:

$$= \rho\,\frac{\partial^2 p}{\partial x^2} + \frac{\partial p}{\partial x}\cdot c\,\rho\,\frac{\partial p}{\partial x} = \rho\,\frac{\partial^2 p}{\partial x^2} + c\,\rho\left(\frac{\partial p}{\partial x}\right)^2$$

Factor out ρ:

$$= \rho\left[\frac{\partial^2 p}{\partial x^2} + c\left(\frac{\partial p}{\partial x}\right)^2\right] \qquad (\,10\,)$$

# Part 10: Put both sides together

Substitute (10) and (9) into (3):

$$\frac{k}{\mu}\,\rho\left[\frac{\partial^2 p}{\partial x^2} + c\left(\frac{\partial p}{\partial x}\right)^2\right] = \phi\,\rho\,c_t\,\frac{\partial p}{\partial t}$$

Divide both sides by ρ (it cancels):

$$\frac{k}{\mu}\left[\frac{\partial^2 p}{\partial x^2} + c\left(\frac{\partial p}{\partial x}\right)^2\right] = \phi\,c_t\,\frac{\partial p}{\partial t}$$

Multiply both sides by μ/k:

$$\frac{\partial^2 p}{\partial x^2} + c\left(\frac{\partial p}{\partial x}\right)^2 = \frac{\phi\,\mu\,c_t}{k}\,\frac{\partial p}{\partial t} \qquad (\,11\,)$$

Equation (11) is **nonlinear** because of the squared gradient term, so it is hard to solve analytically.

# Part 11: Linearization

We compare the two terms on the left of (11):

- Term A: $\partial^2 p/\partial x^2$
- Term B: $c\,(\partial p/\partial x)^2$

For a slightly compressible liquid (oil, water), c is very small, typically $10^{-6}$ to $10^{-5}$ $\text{psi}^{-1}$. The pressure gradient is also small away from the wellbore. Term B contains c times a squared gradient, so it is far smaller than Term A.

**Example:** with $c = 10^{-5}$ $\text{psi}^{-1}$ and $\partial p/\partial x = 10$ psi/ft, Term B is $10^{-5} \times 100 = 10^{-3}$ $\text{psi/ft}^2$. Term A is usually much larger.

So we drop Term B, which gives the diffusivity equation:

$$\frac{\partial^2 p}{\partial x^2} = \frac{\phi\,\mu\,c_t}{k}\frac{\partial p}{\partial t} \qquad (12)$$

# Part 12: Hydraulic diffusivity form

Define

$$\eta = \frac{k}{\phi\,\mu\,c_t}$$

Multiply both sides of (12) by η:

$$\frac{\partial p}{\partial t} = \eta\,\frac{\partial^2 p}{\partial x^2}$$

**SI units check:**

$$\eta = \frac{\text{m}^2}{(\text{Pa}\cdot\text{s})\left(\text{Pa}^{-1}\right)} = \frac{\text{m}^2}{\text{s}}$$

This matches the units of a diffusion coefficient ✓

**Oilfield units:** with k in md, μ in cp, $c_t$ in $\text{psi}^{-1}$, t in hours:

$$\eta = 0.0002637\,\frac{k}{\phi\,\mu\,c_t} \quad \left[\frac{\text{ft}^2}{\text{hr}}\right]$$

# Summary of the logic

| Step | What we did | Result |
|---|---|---|
| 1 | Mass in − mass out = mass accumulated | Balance over Δt |
| 2 | Divide by w·h·Δx·Δt, take limits | Continuity equation (1) |
| 3 | Darcy's law for flow in −x | v = (k/μ) ∂p/∂x |
| 4 | Substitute Darcy into continuity | Equation (3) |
| 5 | Use c and c_f to rewrite ∂(φρ)/∂t | φ ρ c_t ∂p/∂t |
| 6 | Expand ∂(ρ ∂p/∂x)/∂x | Gives the nonlinear c (∂p/∂x)² term |
| 7 | Neglect that term (small c, small gradient) | Linear diffusivity equation |

# Notes: Common Questions and Nuances

## N1. Why c_f for rock and c for fluid?

It is a notation convention. The **f stands for formation** (the rock), not fluid. Fluid compressibility carries a phase subscript in general: c_o (oil), c_w (water), c_g (gas). In the derivation there is a single fluid, so its compressibility is written simply as c. For multiphase flow the total compressibility is

$$c_t = c_f + S_o c_o + S_w c_w + S_g c_g$$

For one liquid phase this reduces to c_t = c + c_f, which is what was used.

## N2. Why is porosity multiplied in the accumulation term?

Fluid is stored only in the pore space. The slice has bulk volume ΔV, but only the fraction φ of it is open pore space:

$$\text{pore volume} = \phi\,\Delta V, \qquad \text{mass stored} = \rho\ (\ \phi\,\Delta V)\ = \Delta V\ (\ \phi\rho\ )$$

Leaving φ out would treat the whole slice as filled with fluid, as if there were no rock.

Two related points:

- **φ is not in the mass in/out terms.** Darcy velocity v is flow rate per unit *total* (bulk) cross-sectional area, so porosity is already built into it.
- **φ sits inside the time derivative, ∂(φρ)/∂t.** Both ρ and φ change with pressure. Fluid expansion changes ρ (compressibility c). Pore volume change

alters φ (compressibility c_f). The product rule on φρ is what produces c + c_f = c_t. Treating φ as constant would lose the rock compressibility.

# 2D and 3D Cartesian Diffusivity Equation

The idea is the same as the 1D case, but fluid now crosses **three pairs of faces**. Here velocities are taken positive in the +x, +y, +z directions, so mass enters at the lower face of each pair and leaves at the upper one. The final result does not depend on this choice.

## 3D Step 1: Control volume and faces

A block with sides Δx, Δy, Δz has volume ΔV = Δx Δy Δz. Faces normal to x have area Δy Δz, faces normal to y have area Δx Δz, faces normal to z have area Δx Δy.

## 3D Step 2: Mass balance over Δt

Mass crossing each face is ρ A v Δt. Writing in minus out for each direction, and the accumulation Δx Δy Δz [$(\varphi\rho)_{t+\Delta t} - (\varphi\rho)_t$]:

$$\begin{aligned}
&\Delta y \Delta z \Delta t\left(\left[\rho v_x\right]_x - \left[\rho v_x\right]_{x+\Delta x}\right) \\
&+\Delta x \Delta z \Delta t\left(\left[\rho v_y\right]_y - \left[\rho v_y\right]_{y+\Delta y}\right) \\
&+\Delta x \Delta y \Delta t\left(\left[\rho v_z\right]_z - \left[\rho v_z\right]_{z+\Delta z}\right) \\
&= \Delta x \Delta y \Delta z\left[\left(\phi\rho\right)_{t+\Delta t} - \left(\phi\rho\right)_t\right]
\end{aligned}$$

## 3D Step 3: Divide and take limits

Divide by Δx Δy Δz Δt. In each term on the left, two lengths cancel and one stays in the denominator. Taking all increments to zero:

$$-\frac{\partial\left(\rho v_x\right)}{\partial x} - \frac{\partial\left(\rho v_y\right)}{\partial y} - \frac{\partial\left(\rho v_z\right)}{\partial z} = \frac{\partial\left(\phi\rho\right)}{\partial t} \qquad (3D\text{-}1)$$

In words: the net mass flowing out of the block in the x, y and z directions equals the decrease of stored mass.

## 3D Step 4: Darcy's law in each direction

$$v_x = -\frac{k}{\mu}\frac{\partial p}{\partial x}, \qquad v_y = -\frac{k}{\mu}\frac{\partial p}{\partial y}, \qquad v_z = -\frac{k}{\mu}\frac{\partial p}{\partial z}$$

Substituting into (3D-1), the two minus signs cancel and k, μ come out of the derivatives:

$$\frac{k}{\mu}\left[\frac{\partial}{\partial x}\left(\rho\frac{\partial p}{\partial x}\right)+\frac{\partial}{\partial y}\left(\rho\frac{\partial p}{\partial y}\right)+\frac{\partial}{\partial z}\left(\rho\frac{\partial p}{\partial z}\right)\right]=\frac{\partial\left(\,\phi\rho\,\right)}{\partial t}$$

## 3D Step 5: Expand and linearize

Each term expands as in 1D (product rule, then ∂ρ/∂x = cρ ∂p/∂x):

$$\frac{\partial}{\partial x}\left(\rho\frac{\partial p}{\partial x}\right)=\rho\frac{\partial^2 p}{\partial x^2}+c\rho\left(\frac{\partial p}{\partial x}\right)^2$$

and similarly for y and z. The accumulation term is unchanged, ∂(φρ)/∂t = φρ c_t ∂p/∂t. Cancelling ρ and multiplying by μ/k:

$$\frac{\partial^2 p}{\partial x^2}+\frac{\partial^2 p}{\partial y^2}+\frac{\partial^2 p}{\partial z^2}+c\left[\left(\frac{\partial p}{\partial x}\right)^2+\left(\frac{\partial p}{\partial y}\right)^2+\left(\frac{\partial p}{\partial z}\right)^2\right]=\frac{\phi\mu c_t}{k}\frac{\partial p}{\partial t}$$

Dropping the small term c[(∂p/∂x)² + (∂p/∂y)² + (∂p/∂z)²] gives the 3D diffusivity equation:

$$\frac{\partial^2 p}{\partial x^2}+\frac{\partial^2 p}{\partial y^2}+\frac{\partial^2 p}{\partial z^2}=\frac{1}{\eta}\frac{\partial p}{\partial t},\qquad \eta=\frac{k}{\phi\mu c_t}$$

## 2D case

For a thin reservoir of thickness h with flow only in the x-y plane, the block is Δx × Δy × h. The thickness h appears in every term and cancels, just as w·h did in 1D. The result is

$$\frac{\partial^2 p}{\partial x^2}+\frac{\partial^2 p}{\partial y^2}=\frac{1}{\eta}\frac{\partial p}{\partial t}$$

## Two practical notes

**Anisotropy.** If permeability differs by direction, Darcy's law uses the matching permeability in each direction:

$$k_x\frac{\partial^2 p}{\partial x^2}+k_y\frac{\partial^2 p}{\partial y^2}+k_z\frac{\partial^2 p}{\partial z^2}=\phi\mu c_t\frac{\partial p}{\partial t}$$

Vertical permeability k_z is usually the smallest.

**Gravity.** In the vertical direction Darcy's law has an extra term, v_z = −(k/μ)(∂p/∂z + ρg) with z measured upward. It was neglected above. For a slightly compressible

liquid, ρ varies very little, so the gravity term is nearly constant in space and drops out of the derivative at this level of approximation. Pressure can then be read as pressure referred to a datum. For accurate vertical-flow problems (gas caps, coning), flow potential is used instead of pressure. The full treatment is in the section Including Gravity.

# Including Gravity

The derivations above neglected gravity. Here it is included, writing every derivative out explicitly. The result is worth stating first:

- For a slightly compressible liquid, **gravity does not change the diffusivity equation**. It changes what the pressure variable means and what the initial condition is.
- The quantity that obeys the diffusivity equation is the **drawdown** p′ = p − p_i(z), where p_i(z) is the hydrostatic pressure before production. Its initial value is zero, and every solution for the no-gravity case applies to it.

The derivation below is written in Cartesian coordinates (x, y, z). Gravity acts only along the vertical direction z, so the steps that involve gravity are the same in any coordinate system. The sections after it show how this applies to the geometry of this PDF.

## Gravity Step 1: Darcy’s law with gravity

Let z be the **elevation, positive upward**, and g the gravitational acceleration. Without gravity the driving force was the pressure slope alone. With gravity, the weight of the fluid adds to the vertical direction only:

$$v_x = -\frac{k}{\mu}\frac{\partial p}{\partial x}, \qquad v_y = -\frac{k}{\mu}\frac{\partial p}{\partial y}, \qquad v_z = -\frac{k}{\mu}\left(\frac{\partial p}{\partial z} + \rho g\right)$$

**Check:** in static equilibrium, pressure increases downward at ∂p/∂z = −ρg, so v_z = 0 ✓.

Horizontal flow does not involve z, so gravity has no effect on it.

## Gravity Step 2: Continuity equation (unchanged)

Gravity does not enter the mass balance, so the continuity equation is the same as in the 3D derivation. Net mass flowing out of a small block in the x, y and z directions equals the decrease of stored mass:

$$-\left[\frac{\partial\left(\rho v_x\right)}{\partial x}+\frac{\partial\left(\rho v_y\right)}{\partial y}+\frac{\partial\left(\rho v_z\right)}{\partial z}\right]=\frac{\partial(\phi\rho)}{\partial t}$$

## Gravity Step 3: Substitute Darcy's law

The minus signs cancel, and k, μ and g are constants. The x and y terms look as before. The z term gains a gravity part, g ∂(ρ²)/∂z, because ρ times the extra term ρg gives ρ²g:

$$\frac{k}{\mu}\left[\frac{\partial}{\partial x}\left(\rho\frac{\partial p}{\partial x}\right)+\frac{\partial}{\partial y}\left(\rho\frac{\partial p}{\partial y}\right)+\frac{\partial}{\partial z}\left(\rho\frac{\partial p}{\partial z}\right)+g\frac{\partial\left(\rho^2\right)}{\partial z}\right]=\frac{\partial(\phi\rho)}{\partial t}$$

## Gravity Step 4: Expand each term

**x, y and z pressure terms** (product rule, with ∂ρ/∂x = cρ ∂p/∂x and so on). For x:

$$\frac{\partial}{\partial x}\left(\rho\frac{\partial p}{\partial x}\right)=\rho\frac{\partial^2 p}{\partial x^2}+c\rho\left(\frac{\partial p}{\partial x}\right)^2$$

The y and z terms have the same form.

**Gravity part.** Use ∂(ρ²)/∂z = 2ρ ∂ρ/∂z and ∂ρ/∂z = cρ ∂p/∂z:

$$g\frac{\partial\left(\rho^2\right)}{\partial z}=2c\rho^2 g\frac{\partial p}{\partial z}$$

**Accumulation part** (unchanged): ∂(φρ)/∂t = φρ c_t ∂p/∂t.

## Gravity Step 5: Combine, cancel ρ, multiply by μ/k

$$\frac{\partial^2 p}{\partial x^2}+\frac{\partial^2 p}{\partial y^2}+\frac{\partial^2 p}{\partial z^2}+c\left[\left(\frac{\partial p}{\partial x}\right)^2+\left(\frac{\partial p}{\partial y}\right)^2+\left(\frac{\partial p}{\partial z}\right)^2\right]+2c\rho g\frac{\partial p}{\partial z}=\frac{1}{\eta}\frac{\partial p}{\partial t}\qquad(G1)$$

Compared with the no-gravity case, there is one extra term, 2cρg ∂p/∂z.

## Gravity Step 6: The hydrostatic initial state

Before production there is no flow, so v_z = 0 and ∂p/∂z = −ρg. To first order (ρ constant over the height of the reservoir):

$$p_i\left(z\right)=p_{ref}-\rho g\left(z-z_{ref}\right)$$

Pressure is therefore **not uniform** initially. It varies with elevation. Define the drawdown as the departure from this state:

$$p'\left(x,y,z,t\right)=p-p_i\left(z\right),\qquad p'\left(x,y,z,0\right)=0$$

## Gravity Step 7: Substitute the drawdown into (G1)

The hydrostatic part depends on z only, so the x and y slopes are unchanged, and ∂p_i/∂z = −ρg gives

$$\frac{\partial p}{\partial x} = \frac{\partial p'}{\partial x}, \qquad \frac{\partial p}{\partial y} = \frac{\partial p'}{\partial y}, \qquad \frac{\partial p}{\partial z} = \frac{\partial p'}{\partial z} - \rho g$$

For the second derivatives and the squared slope in z:

$$\frac{\partial^2 p}{\partial z^2} = \frac{\partial^2 p'}{\partial z^2} + \frac{d^2 p_i}{dz^2}, \qquad \left(\frac{\partial p}{\partial z}\right)^2 = \left(\frac{\partial p'}{\partial z}\right)^2 - 2\rho g \frac{\partial p'}{\partial z} + \left(\rho g\right)^2$$

Also ∂p/∂t = ∂p′/∂t, because p_i does not depend on time. The x and y terms of (G1) are unchanged apart from p becoming p′. The z terms of (G1) become:

$$\begin{aligned} &\tfrac{\partial^2 p'}{\partial z^2} + \tfrac{d^2 p_i}{dz^2} + c\left(\tfrac{\partial p'}{\partial z}\right)^2 - 2c\rho g \tfrac{\partial p'}{\partial z} + c\left(\rho g\right)^2 \\ &\quad + 2c\rho g \tfrac{\partial p'}{\partial z} - 2c\left(\rho g\right)^2 \end{aligned}$$

Two cancellations happen:

1. **The cross terms** −2cρg ∂p′/∂z and +2cρg ∂p′/∂z cancel each other.
2. **The static terms.** Differentiating the hydrostatic condition dp_i/dz = −ρg once more gives d²p_i/dz² = −g dρ/dz = c(ρg)². So d²p_i/dz² + c(ρg)² − 2c(ρg)² = 0.

What remains is exactly the no-gravity equation, written for the drawdown:

$$\frac{\partial^2 p'}{\partial x^2} + \frac{\partial^2 p'}{\partial y^2} + \frac{\partial^2 p'}{\partial z^2} + c\left[\left(\frac{\partial p'}{\partial x}\right)^2 + \left(\frac{\partial p'}{\partial y}\right)^2 + \left(\frac{\partial p'}{\partial z}\right)^2\right] = \frac{1}{\eta}\frac{\partial p'}{\partial t}$$

(The density in the gravity term is the local density, which differs from the hydrostatic density by a relative amount of order c·p′, about $10^{-4}$. The cancellation is accurate to that order.)

Now linearize as before, by dropping the small bracket multiplied by c:

$$\frac{\partial^2 p'}{\partial x^2} + \frac{\partial^2 p'}{\partial y^2} + \frac{\partial^2 p'}{\partial z^2} = \frac{1}{\eta}\frac{\partial p'}{\partial t}, \qquad p'\left(x, y, z, 0\right) = 0$$

## Gravity Step 8: Equivalent view with flow potential

Define the flow potential Φ = p + ρ g z. For constant ρ, Darcy's law becomes

$$v_x = -\frac{k}{\mu}\frac{\partial \Phi}{\partial x}, \qquad v_y = -\frac{k}{\mu}\frac{\partial \Phi}{\partial y}, \qquad v_z = -\frac{k}{\mu}\frac{\partial \Phi}{\partial z}$$

At equilibrium Φ is uniform:

$$\Phi_i = p_{ref} + \rho g\, z_{ref}$$

and Φ − Φ_i equals p′. So Φ obeys the diffusivity equation with a **uniform** initial value, which is why potential is often used for problems with significant vertical flow.

**In practice:** solve for p′ with the formulas from the no-gravity derivation, then add back the hydrostatic part to get the pressure at a given elevation: p = p_i(z) + p′.

## Gravity Step 9: An independent size check

Even without the cancellation above, the gravity term is small. Take water with ρg ≈ 0.433 psi/ft and c ≈ 3 × 10⁻⁶ psi⁻¹:

$$2c\ (\ \rho g\ )^{\,2} \approx 2\ \left(\ 3\times 10^{-6}\ \right)\ \ (\ 0.433\ )^{\,2} \approx 1.1\times 10^{-6}\ \mathrm{psi/ft}^{2}$$

A typical diffusion term for a 100 psi drawdown over 100 ft is ∂²p/∂z² ≈ 100/100² = 0.01 psi/ft². The gravity term is about 10⁻⁴ of that, the same negligible order as the c(slope)² terms already dropped.

## Gravity in Linear Flow

**Horizontal linear flow.** The elevation z is constant along the flow path, so ∂z/∂x = 0 and there is no gravity term. Every result in this PDF applies as written.

**Dipping linear flow.** Let x be measured up-dip along a bed with dip angle α, so the elevation is z = x sinα. Darcy's law along x picks up the component of gravity along the bed:

$$v = \ -\frac{k}{\mu}\left(\frac{\partial p}{\partial x} + \rho g\,\sin\,\alpha\right)$$

For flow in the −x direction, as in the derivation, the sign of v flips. Substituting into the continuity equation ∂(ρv)/∂x = ∂(φρ)/∂t gives the same structure as Step 3:

$$\frac{k}{\mu}\left[\frac{\partial}{\partial x}\left(\rho\frac{\partial p}{\partial x}\right) + g\,\sin\,\alpha\,\frac{\partial\,(\ \rho^{2}\ )}{\partial x}\right] = \frac{\partial\,(\ \phi\rho\ )}{\partial t}$$

The gravity term is 2cρ² g sinα ∂p/∂x. With the hydrostatic state p_i(x) = p_ref − ρ g sinα (x − x_ref), exactly the same cancellation as in Step 7 occurs, and the drawdown obeys

$$\frac{\partial^{2} p'}{\partial x^{2}} = \frac{1}{\eta}\frac{\partial p'}{\partial t}$$

## When gravity cannot be dropped

1. **Gas.** Density varies strongly with pressure, so the hydrostatic gradient is not constant and the term $2c\rho g\, \partial p/\partial z$ is not small. Pseudo-pressure with a gas-gradient correction is used.
2. **Multiphase flow.** Oil, gas and water have different densities, so gravity segregation drives vertical flow (coning, gas-cap expansion, gravity drainage). Each phase has its own Darcy's law with its own $\rho g$.
3. **Strongly dipping or very thick reservoirs.** The potential difference across the pay zone can matter in water-drive and gravity-drainage problems.
4. **Datum conversion.** When you compare wells at different depths, convert all pressures to a common datum using the fluid gradient. In field units the gradient is $0.433 \cdot \gamma$ psi/ft, where $\gamma$ is the fluid specific gravity.